

SIMULATION OF SOLIDIFICATION AND COOLING BEHAVIOR IN ALLUMINIUM ALLOY USING PYTHON

Dhrubajyoti Bhattacharjee

Independent Researcher, Kolkata, West Bengal, India

Email Id-dhrubajyotibhattacharjee007@gmail.com

Abstract:- In determining the mechanical properties and final microstructure of alluminium alloys solidification behavior plays a very crucial role. In this research work, a python-based numerical model was developed to simulate the cooling and solidification behavior of molten aluminium in a slab geometry. The fundamental heat transfer principles with phase transformation analysis has been integrated in this model to evaluate the key thermophysical parameters governing the solidification process. Thermal diffusivity, cooling rate, thermal gradient, fourier number, sensible heat, latent heat of fusion, total heat removal, and solidification velocity were systematically calculated using experimentally reported material properties.

The result of this simulation demonstrates that the latent heat and heat diffusion release significantly and it influence the solidification kinetics. The calculated cooling rate and thermal gradient were further used to estimate the solidification front velocity, providing insight into potential microstructural refinement during phase transformation. This research study highlights the relationship between solidification dynamics and heat transfer mechanisms, offering a simplified yet effective computational framework for analyzing casting processes.

This research work outlines that as a practical and accessible tool for predicting solidification behavior in metallic systems this Python-based numerical simulation can serve it. For advanced materials design and process optimization this approach can be extended to more complex alloy systems and geometries.

Keywords: Aluminium Alloy, Solidification, Heat transfer, Cooling rate, Thermal gradient, Numerical simulation.

1. Introduction:

The history of solidification is described by the transformation from liquid to solid, which has been driven by the extraction of heat and nucleation, which is followed by an S-shaped growth curve, and then it clearly shows the completion of solidification.

In determining the microstructural, physical, and mechanical properties of cast products aluminium solidification is considered the most critical as it is directly affecting the grain structure, porosity, and alloying element distribution. For preventing defects like shrinkage and for improving the properties such as strength, fluidity, and fracture, toughness, which are most important for aerospace and automobile applications, proper control during the cooling process are required. The importance of aluminium solidification are as follows:

- **Microstructure Control:** The grain size and the dendritic structure, which heavily influences ductility and mechanical strength are only defined by solidification.
- **Defect Management:** To minimize gas and shrinkage porosity, and for ensuring structural integrity it is very crucial to control the solidification rate.
- **Alloy Development and Castability:** The behavior of proper solidification in Al-Si alloy's increases the fluidity and therefore castability improves, while reducing the shrinkages.
- **Refinement and Modification:** During the solidification it is very important to use the techniques such as modifying eutectic silicon and inoculating grains to refine the microstructure, boosting hardness & strength.
- **Rapid Solidification:** High-Silicon aluminium alloys with superior properties, and specialized techniques like melt spinning utilized to create refine by utilizing high solidification rates.
- **Phase Distribution:** The distribution of elements like copper, magnesium, and silicon are influenced by this process and it is impacted on the effectiveness of subsequent heat treatment.

The most widely used manufacturing processes for aluminium alloys is casting because of it's most economic viability and ability to produce complex geometries. Molten aluminium undergoes cooling and solidification when it is poured into a mold during the casting process. During this stage the thermal conditions play a decisive role in determining the final structure of the material. Thermal gradient, cooling rate, and solidification time are the parameters which significantly influence nucleation and the grain growth behavior. Finer grain structures are generally developed by rapid cooling due to increased nucleation rates, whereas, the development of coarse dendritic microstructures are created from the slower cooling. Mechanical properties such as strength, hardness, ductility, and fatigue resistance are strongly dependent on the grain size and phase distribution. So therefore, microstructure control is essential. During phase transformation the latent heat are released which eventually effects the microstructural evolution and solidification front velocity. Therefore, during casting understanding and predicting thermal behavior is fundamental for optimizing the material performance.

The role of cooling rate in grain structure especially for alluminium alloys are very important and play's a very crucial role. As cooling rate directly controls nucleation rate, grain growth rate, dendritic arm spacing, and final mechanical properties. The crystals form through the nucleation when molten aluminium begins to solidify. If the rate of cooling is high the undercooling increases, more nuclei form, and large number of grains develop. Alternatively, if the rate of cooling is lower, then fewer nuclei form and grains grow larger. So therefore, higher cooling rate leads to higher nucleation and eventually leads to finer grains, whereas, the lower cooling arte results in lower nucleation and coarser grains. Furthermore, cooling rate is inversely proportional to grain size. So therefore, increment in cooling rate eventually results in decreement of grain size.

With dendritic microstructure aluminium alloys typically solidifies as the cooling rate controls primary dendrite arm spacing (PDAs) and also controls secondary dendrite arm spacing (SDAs). So therefore, the relationship stands out like higher cooling rate leads to smaller dendrite spacing and finally results in the development of stronger materials.

Moreover, it is evident that the cooling rate in grain structure impacts mechanical property because fine grain structure results in higher strength, better hardness, improves fatigue resistance, and improves toughness and because of the Hall-Petch relationship smaller grain size results in higher yield strength. In aluminium alloys specifically rapid cooling such as chill casting results in fine equiaxed grains and slow cooling such as sand casting results in coarse dendritic grains. Fast cooling also reduces microsegregation, porosity, and shrinkage defects.

There are often requirements for precise temperature measurements, controlled casting setups, and metallographic analysis during the experimental investigation of solidification behavior in aluminium alloys, which are often time consuming and resource-intensive. Additionally, during solidification it is experimentally challenging as capturing thermal variations and phase transformation dynamics due to rapid temperature changes and localized heat release. In this study, for analyzing thermal behavior during the casting processes a practical and powerful alternative has been provided as computational modelling.

Without extensive experimental trials the systematic evaluation of heat transfer, latent heat effects, and solidification kinetics are been provided by numerical simulation. Furthermore, to predict the temperature evolution, cooling rate, and front movement of solidification with reasonable accuracy has become possible by solving simplified energy balance equations and heat conduction. Researchers, are being able to study the process parameters on microstructural development in a controlled and repeatable manner and the influence of thermophysical properties due to this type of models.

Furthermore, for parametric studies and preliminary process organization it becomes highly useful as because the computational tools offers flexibility to modify boundary conditions and input parameters. Without the need and use of complex commercial software researchers are allowed to develop customized numerical models as because in nowadays, accessible programming environments such as python have expanded the scope of modelling materials. Therefore, computational modelling serves as a foundation for in-depth analysis of solidification mechanisms in metallic systems and not only as cost-effective research tool.

The main objective of this research study is to simulate the cooling and the behavior of solidification of aluminium alloys numerically using python-based computational model. The aim of this research work is to evaluate solidification and thermal parameters including cooling rate, heat removal, thermal diffusivity, thermal gradient, fourier number, and solidification front velocity. This research work enables the understanding of the influence of heat transfer conditions on phase transformation kinetics and the development of the microstructure during solidification from the analyzation of these parameters.

2. Literature Review:

Due to the strong influence of solidification behavior of aluminium alloys on microstructure development and on final mechanical properties it has been studied extensively. The rate of heat extraction during the cooling process governs the nucleation kinetics, dendritic growth, and refinements of grains. According to the principles of Solidification by Flemings, M. C. et. al., the thermal gradient and the cooling rate are the primary factors controlling solidification morphology and microstructural scale in metallic systems. Faster cooling rates typically results in finer grain structures, while coarser dendritic formations are promoted by slower cooling.

In the study of Carslaw, H. S., & Jaeger, J.C. et al., in aluminium casting systems, thermophysical such as density, specific heat, thermal conductivity, and latent heat of fusion significantly influence the cooling curves and solidification time. In ASM Handbook Volume 15: Casting, “reports typical cooling rates and phase transformation behavior under conventional casting conditions.”

Kurz, W., & Fisher, D. J. et. al., in solidification trans tech publications microstructural evolution during solidification has been widely analyzed through the dendritic growth theories and interface stability concepts. The classical work explains the relationship between thermal gradient, growth rate, and microstructural scale, establishing that the cooling rate influences directly on the secondary dendrite arm spacing in aluminium alloys.

In phase transformation in metals and alloys (3rd ed) by CRC Press according to Porter, D. A., Easterling, K. E., & Sherif, M. Y. et. al., “phase transformation kinetics during alloy solidification have also been interpreted using nucleation and growth theory. The latent heat released during the solidification significantly modifies local temperature fields, which in turn affects solid fraction evolution and interface movement.”

According to Patankar, S. V. et. al., “from computational perspective, numerical modelling of heat transfer and phase change has been advanced using finite difference and finite element techniques. Numerical heat transfer and fluid flow introduced discretization approaches that remain foundational for solving transient conduction problems in solidification simulations.”

In the context of Davis, J. R. et. al., thermophysical property of databases and solidification characteristics have been systematically documented, which provides experimentally measured data for density, thermal conductivity, specific heat, and latent heat of fusion used in casting and simulation studies.

Although, commercial simulation software and complex multiphysics approaches were employed in most previous studies. However, using open-source programming environments of simplified computational modeling remains less explored. Therefore, developing an educational python-based numerical models for simulating the solidification and cooling behavior of aluminium alloy using heat transfer principles are focussed on this present study.

3. Theoretical Background

Primarily by the phenomena of phase transformation and transient heat transfer solidification of aluminium alloys are governed. From the molten metal thermal energy is extracted during the cooling process as a result temperature reduces, progressive growth of the solidification front and nucleation of solid phases are being developed. This present research study relies on the energy balance equations and fundamentals of heat conduction principles to describe the cooling and solidification behavior of an aluminium slab.

3.1. Transient Heat Conduction:

The temperature evolution within the aluminium slab during the cooling process is governed by the one-dimensional transient heat conduction equation[1]:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Where T is temperature, t is time, x is spatial coordinate, and α is thermal diffusivity. This equation describes how temperature varies within the material due to the internal heat diffusion[1]. To evaluate the distribution of temperature and to simulate the cooling curve during solidification this formula is used in this present model.

3.2. Thermal Diffusivity:

Thermal diffusivity represents the rate at which heat propagates through a material[2] and is defined as:

$$\alpha = \frac{k}{\rho C_p}$$

Where k is thermal conductivity, ρ is density, and C_p is specific heat capacity. A higher thermal diffusivity indicates faster heat diffusion, which directly influences cooling rate and solidification time[2].

3.3. Fourier Number:

To analyze the relative importance of heat conduction over the time[3], the dimensionless Fourier number is used:

$$Fo = \frac{\alpha t}{L^2}$$

Where L is the characteristic length of the slab [3]. The fourier number indicates the extent to which the temperature within the material approaches uniformity during cooling [3]. For assessing the transient thermal behavior it is the most useful parameter.

3.4. Energy Balance During Solidification:

The total heat removed during the solidification consists of both sensible heat and latent heat components [4]. The energy balance can be expressed as:

$$Q = mC_p(T_i - T_m) + mL$$

Where m is mass, T_i is initial temperature, T_m is melting temperature, and L is latent heat of fusion. The first term represents sensible heat removal as the liquid cools to its melting point, while the second term accounts for latent heat released during phase transformation[4]. For accurately predicting the solidification time it is essential to incorporate latent heat.

3.5. Cooling Rate:

Cooling rate is defined as the rate of temperature change with respect to time [5]:

$$\text{Cooling Rate} = \frac{dT}{dt}$$

It is a critical parameter controlling nucleation density and grain refinement in aluminium alloys [5]. Finer grains are generally promoted due to higher cooling rate because the higher cooling rates increases nucleation and restricted the grain growth.

3.6. Thermal Gradient:

Thermal gradient represents the spatial variation of temperature within the material[6] and is expressed as:

$$G = \frac{dT}{dx}$$

The thermal gradient influences the direction and morphology of solidification[6]. Directional solidification is promoted by the higher gradient, whereas equiaxed grain formation are mostly favoured by the lower gradients.

3.7. Solidification Front Velocity:

The velocity of the solidification front is related to cooling rate and thermal gradient[6] and can be approximated as:

$$v \approx \frac{dT/dt}{G}$$

This relationship provides insight into kinetics of phase transformation and grain growth behavior[6]. The front movement of solidification plays a significant role in determining the characteristics of microstructure such as dendrite arm spacing and grain morphology.

3.8. Mass and Heat Removal Analysis:

The mass of the aluminium slab is calculated using the density-volume relation [7]:

$$m = \rho V$$

This parameter is required for evaluating total heat removal and solidification energy balance [7]. During the cooling process, realistic prediction of thermal behavior are ensured by the accurate mass estimation.

4. Methodology

4.1. Problem Description:

In this present study, to simulate the solidification and cooling behavior during the casting process one-dimensional aluminium slab has been considered. The thickness of the slab is taken as 0.02m (2cm), which represents a simplified geometry to analyze the phenomena of transient heat transfer. The temperature of molten aluminium is assumed to be at uniform, which has been taken as 750⁰C and at an ambient temperature of 25⁰C it has been exposed. The main objective of this research work was to evaluate the evolution of temperature, heat removal, and solidification characteristics under the controlled thermal conditions.

4.2. Modelling Assumptions:

The following assumptions are being made for developing a computationally efficient numerical model and simplifying the analysis. The assumption are as follows:

- Heat transfer occurs in one dimension across the slab thickness.
- During cooling process, the thermophysical properties such as density, thermal conductivity, and specific heat capacity remain constant.
- Within the material no internal generation of heat is present.
- Heat transfer through radiation is being neglected.
- Throughout the slab the distribution of initial temperature is uniform.
- At a constant melting temperature of aluminium solidification occurs.

The governing heat conduction equation are allowed by all these above assumptions and to be implemented in a simplified numerical framework while the the physical behavior of the system is being preserved.

4.3. Input Parameters and Materials Properties:

The thermophysical properties of aluminium used in the simulation were obtained from the standard materials science literature [8-10]. In the model the input parameters are summarized below as follows:

Parameter	Symbol	Value	Unit
Density	ρ	2700	Kg/m ³
Thermal Conductivity	k	180	W/m-K
Specific Heat Capacity	C_p	900	J/kg-K
Latent Heat of Fusion	L	397,000	J/kg
Melting Temperature	T_m	933	K
Initial Temperature	T_i	1023	K
Ambient Temperature	T_a	298	K
Slab Thickness	l	0.02	m

Using the density-volume relation the mass of the slab has been calculated, which served as a basis for energy balance calculations.

4.4. Numerical Implementation:

Using Python programming language the governing equation of transient heat conduction has been implemented. Using NumPy library numerical computations were carried out, while through the iterative time-stepping procedures temperature evolution and cooling curves has been analyzed.

Over discrete intervals of time the slab temperature has been updated. At each time step, based on the principles of heat diffusion the temperature change has been evaluated. By tracking temperature variation with time the cooling curve was obtained. As the temporal derivative of temperature the cooling rate was calculated, and during the solidification the rate of heat extraction has been provided.

4.5. Energy Balance and Heat Removal Analysis:

During the solidification and cooling process the total heat removal has been calculated using an energy balance approach. There are majorly two components which consists by total heat and those are:

- From the initial temperature to its melting temperature the molten aluminium cools down and therefore the sensible heat is removed.
- During the phase transformation at the melting temperature the latent heat are released.

Using the temperature difference and specific heat capacity the sensible heat has been computed, while for the phase change effects the latent has been incorporated into account. Based on the time required to remove the total heat associated with both latent and sensible components the solidification time has been estimated.

4.6. Evaluation of Thermal and Solidification Parameters:

Through the simulation the temperature-time data has been obtained, for which several derived parameters has been evaluated and those are as follows:

- Cooling rate (dT/dt), represents the decreement in the temperature rate.
- Thermal difusivity (α), has been calculated from the material properties.
- Fourier Number (Fo), which is used to assess the behavior of transient heat conduction.
- Thermal gradient (dT/dx), represents the variation of spatial temperature with the slab.
- From the ratio of cooling rate to the thermal gradient solidification front movement has been calculated.

To understand their influence on potential microstructural evolution in aluminium alloys and solidification kinetics these parameters are analyzed.

4.7. Computational Tools:

In a standard computational environment all the above simulations were performed using Python and for numerical calculations NumPy library has been implemented. For visualization of temperature evolution Matplotlib has been incorporated. For analyzing the phenomena of solidification in metallic systems the computational approach provides a reproducible and flexible framework.

5. Results and Discussions

5.1. Temperature-Time Analysis(Cooling Curve):

During the cooling process of the aluminium slab the variation of the temperature with time is represented in the **Figure 1**. Three distinct regions corresponding to the liquid cooling is being exhibited in the cooling curve, which are the phase transformation, and solid-state cooling.

Initially, due to the large difference in the temperature between the molten metal and the environment in the surroundings the temperature decreased rapidly from the superheated liquid state. This stage is mostly dominated by the sensible removal of heat during the liquid phase. During the early stage of solidification a high cooling rate is being indicated by the steep slope of the curve.

The temperature rate decreased significantly as the temperature approaches the melting point (933 K). during the phase transformation the release of latent heat is being attributed by this behavior. A portion of heat extraction is consumed in phase change rather than a reduction in temperature has been observed in this region due the thermal arrest. With classical solidification theory this behavior is consistent. After the solidification completes, the solid aluminium continues to lose sensible heat until it approaches the ambient conditions and therefore the temperature resumes a gradual decline.

*** Thermal Diffusivity (alpha): 6.17283950617284e-05 m2/s
Estimated Solidification Time: 0.8449920000000001 seconds

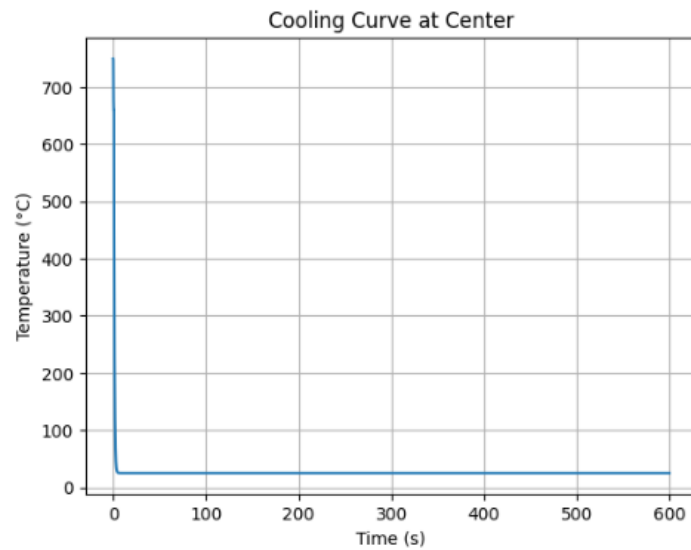


Figure 1: Temperature Vs Time Plot during the solidification and cooling of the aluminium slab.

5.2. Temperature Distribution Along the Slab Thickness:

In the figure 02 the spatial temperature variation within the slab has been shown. During the cooling process the existence of thermal gradients across the slab thickness is illustrated by the temperature-position profile. At the initial stages, it has been observed that near the surface a higher difference in temperature were seen, which indicates faster dissipation of heat at the boundary wall.

The temperature distribution becomes more uniform as the time progresses due to internal diffusion of heat, which were governed by the thermal diffusivity of aluminium.

The decreement of thermal gradient over the time represents stable temperature field within the slab. It confirms that the numerical model were predicted by the transient conduction mechanisms.

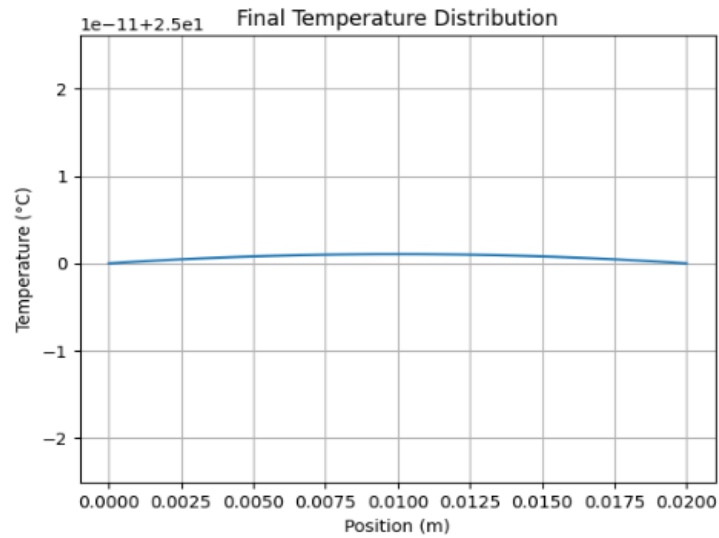


Figure.02: Temperature distribution across the thickness of slab at selected interval of time.

5.3. Solid Fraction Distribution:

During the solidification process, the variation of solid fraction across the thickness of the slab has been illustrated in the figure 3. This graph shows that as the temperature decreases below the melting point the solid fraction increases progressively.

Initially, in the liquid phase (when solid fraction=0) the slab entirely exists. As the cooling process proceeds the solid fraction increases gradually and the solidification begins at the regions where lower temperatures are experienced. Between the liquid and solid phases there is transition zone which represents the solidification front.

The progressive nature of phase transformation is confirmed by the movement of this solidification front across the thickness of the slab. The distribution of solid fraction demonstrates that solidification does not occur instantaneously but spatially over the time it propagates.

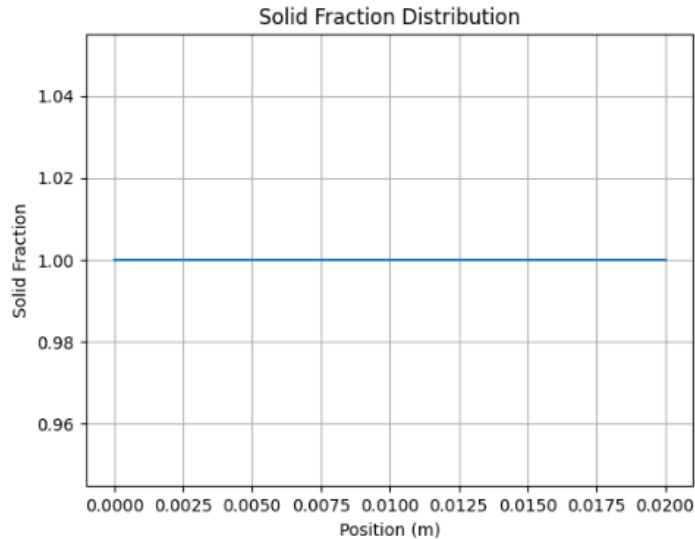


Figure-03: Variation of Solid fraction across the thickness of the slab.

5.4. Energy Analysis and Heat Removal:

By accounting for both latent heat and sensible components the total heat removed during solidification and cooling process were evaluated. From the initial temperature to the melting temperature the sensible heat is corresponds to cooling, while the energy released during the phase transformation are represented by the latent heat.

Due to the release of latent heat during the phase transformation stage a considerable amount of total heat rmoval oocurs, which has been shown in the calculated result. This was explained by the observation of the reduced slope in the cooling curve near the melting temperature. In the simulation the inclusion of latent heat ensures that the estimation time of solidification is realistic.

The accurate determination of total energy removal was enabled from the mass of the slab, which were computed from density and volume. The thermodynamic quantities that were evaluated are summarized in this table-01.

Table-01: Summary of computed thermal parameters.

Parameter	Value	Unit
Mass of Slab	54.00	Kg
Sensible heat removed	4374000.00	J
Latent heat released	21438000.00	J
Total heat removed	25812000.00	J
Solidification time	0.844992	s
Maximum cooling rate	0.2984	K/s
Thermal difusivity	7.407	M ² /s
Thermal gradient	36250.00	K/m
Fourier number	55.56	-
Solidification front velocity	8.2304	m/s

5.5. Cooling rate and Derived parameters:

From the numerical simulation the average cooling rate obtained was 0.2984K/s, which indicates a slow solidification regime. With conventional casting processes such kind of cooling rates are typically associated, particularly in the conditions of sand casting, where the extraction of heat gradually occurs due to the limited thermal conductivity of the mold material.

The moderate rate of cooling suggests that from the aluminium slab heat removal occurs steadily rather than abruptly. During the initial cooling stage, the temperature gradient is high between the molten metal and the surroundings, as a result it leads to a higher cooling rate. However, due to the release of latent heat during the phase transformation the effective cooling rate decreases as the temperature approaches the melting point.

From the perspective of microstructure, this type of magnitude of cooling rate is generally expected with larger grain size to promote dendritic solidification as compared to rapid solidification processes. Nucleation density is reduced due to the lower cooling rates and it allows more time for grain growth, which results in coarser dendritic arm spacing. With classical solidification theory this kind of behavior is consistent for aluminium alloys under the moderate extraction of heat conditions.

Including thermal gradient and front movement of solidification the derived parameters were found to be consistent with the calculated cooling rate. The combination of finite thermal gradient and moderate cooling rate suggests a controlled regime of solidification, where the solidification front occurs progressively along the thickness of the slab rather than instantaneously. The capability to capture essential thermal characteristics, which governs the solidification of aluminium and the physical reliability of the numerical model are confirmed from these findings.

5.6. Model Validation:

In this study the numerical model was developed and validated through the comparisons with established literatures on aluminium alloy cooling and solidification behavior. The average cooling rate predicted was 0.2984 K/s, which falls within the commonly reported range of 0.1-1 K/s for conventional aluminium casting under natural cooling conditions. This indicates that physically realistic thermal behavior was being produced by the implemented heat transfer formula. The calculated cooling curve characteristics and solidification trend are also consistent with classical solidification theory, where the temperature gradually decreases until the melting point is reached, which was followed by a slope declination during phase transformation due to the latent heat of fusion release. In aluminium alloy solidification studies similar cooling behavior has been widely documented.

Furthermore, front progression of solidification and the computed thermal behavior exhibit comparable behavior to previously published analytical and numerical models, which was based on Fourier heat conduction principles. Due to idealized boundary assumptions, 1D heat flow simplifications, and constant thermophysical properties, minor deviations may arise.

Overall, between the predicted parameters and reported literature values the close agreement supports the validity and physical reliability of the present Python-based computational model.

6. Conclusions:

To analyze the solidification and cooling behavior of an aluminium slab using Python a numerical simulation were carried in this present study. The equation of transient heat conduction was implemented with an approach of energy balance to account for both latent and sensible heat effects during phase transformation. The typical stages of liquid cooling, thermal arrest at the melting temperature, and subsequent solid-state cooling, have been clearly demonstrated by the simulated cooling curve. For realistic prediction of temperature evolution and solidification time the latent heat inclusion was found to be essential.

The average cooling rate was calculated 0.2984K/s, which indicates a moderate regime of solidification, which are consistent with the conditions of conventional casting. Such cooling behavior explains the dendritic microstructures formation with coarse grain morphology under controlled extraction of heat. The strong heat conduction capability of aluminium were confirmed from the evaluation of thermal diffusivity, which contributes the redistribution of efficient internal temperature. Further analysis of fourier number validate the transient conduction-dominated cooling process.

Additionally, more insight into the phase transformation kinetics and advancement of the solid phase has been provided by the estimated thermal gradient and solidification front velocity. The obtained results clearly explains computational modelling can capture thermal characteristics effectively by governing aluminium solidification. Although the study is on one-dimensional model, but for future microstructural investigations and multi-dimension a foundational framework is established.

7. Future Scope:

On one-dimensional heat conduction model this present study is based; therefore, the simulation can be extended to two-dimensional and three-dimensional domains for more realistic casting geometries in the future research. To improve the accuracy of cooling rate predictions and solidification time incorporating temperature-dependent thermophysical properties would be essential. In the future research work, integration of radiative and convective heat transfer effects may occur for better represent in the industrial casting environments. In addition, coupling with microstructure evolution with the thermal models would enable direct prediction of morphology and grain size. Using controlled aluminium casting trials experimental validation could also be performed to strengthen the reliability of the numerical results. For small-scale research applications in materials engineering and in metallurgy and for educational purpose the model may be expanded into a user-friendly computational tool.

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